

$^{39}\text{K}(\gamma, \alpha)$ 1987Ar08

1987Ar08: $E_{\text{max}}=32\text{--MeV}$ γ -ray beam was produced from the betatron at Moscow University Nuclear Physics Research Institute and impinged on a natural metallic potassium target of mass $\approx 200\text{ g}$. Photons from (γ, p) , (γ, n) , and (γ, α) from the giant dipole resonance (GDR) in ^{39}K were detected using a 100-cm^3 Ge(Li) detector at 140° . Measured E_γ . Deduced partial photonuclear cross sections, levels, J , π .

 ^{35}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>Comments</u>
0	$3/2^+$	
1760	$5/2^+$	$T=1/2$

† As given in 1987Ar08.

‡ From Adopted Levels.

 $\gamma(^{35}\text{Cl})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1760	1760	$5/2^+$	0	$3/2^+$

 $^{39}\text{K}(\gamma, \alpha)$ 1987Ar08Level Scheme